Project Synopsis: ESP32-Based Air Quality and Weather Monitoring System

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Abstract

This project builds a low-cost IoT system that monitors air quality, temperature and humidity in real time. An ESP32 board reads an MQ135 gas sensor and a DHT22 sensor, shows the values on a 0.96-inch OLED display, and sends them over Wi-Fi to a ThingSpeak cloud dashboard. The dashboard plots current and historical readings as graphs, so conditions can be tracked remotely and trends can be analysed. The system labels air quality as Good, Moderate or Poor, and uses only commonly available components, which makes it suitable for classrooms, laboratories, homes and small workplaces.

Introduction

Air pollution and changing environmental conditions affect human health, indoor comfort and workplace safety. Keeping track of air quality, temperature and humidity helps people notice unhealthy conditions early and act on them.

This project combines an ESP32 microcontroller with two low-cost sensors and an OLED display to measure these conditions continuously. Because the ESP32 has built-in Wi-Fi, the same readings can also be pushed to a cloud platform, giving remote access and a history of past values without extra hardware.

Problem Statement

Traditional air quality monitoring systems are often expensive and may not be suitable for small-scale applications such as a classroom, a laboratory or a small workplace.

There is a need for a simple and affordable solution that measures air quality and weather parameters using easily available electronic components, shows them locally, and makes them viewable remotely.

Objectives

- Design a low-cost system that monitors air quality, temperature and humidity in real time.
- Display live readings locally on an OLED screen.
- Send readings over Wi-Fi to a cloud dashboard for remote monitoring.
- Visualise historical data as graphs to show trends.
- Classify air quality as Good, Moderate or Poor for quick interpretation.
- Keep the design expandable so more environmental sensors can be added later.

Scope of the Project

The project covers sensing, local display, Wi-Fi transfer and cloud visualisation for indoor and small-scale environments such as classrooms, laboratories, homes and small workplaces.

It is a demonstration and learning system. It is not a replacement for certified industrial air quality monitoring equipment, and its gas readings are indicative until the MQ135 sensor is calibrated.

Proposed System and Methodology

The ESP32 is the controller. It reads both sensors, shows the values on the OLED, and uploads them to ThingSpeak through Wi-Fi. The system works as follows:

1. The MQ135 sensor detects changes in air-quality-related gas concentration.
2. The DHT22 sensor measures temperature and humidity.
3. The ESP32 reads and processes the sensor values.
4. The OLED display shows the live readings.
5. The ESP32 sends the readings to a cloud dashboard through Wi-Fi.
6. The dashboard displays current and historical data using graphs.
7. Users observe environmental changes and identify unhealthy conditions.

The firmware is a single Arduino sketch, `esp32_air_quality_monitor.ino`. The ThingSpeak channel uses three fields: air quality value, temperature and humidity. Serial output at 115200 baud is available for checking readings during testing.

Hardware and Software Requirements

Hardware

Component	Quantity	Purpose
ESP32 Development Board	1	Main controller and Wi-Fi module
MQ135 Gas Sensor	1	Air quality sensing
DHT22 Sensor	1	Temperature and humidity sensing
OLED Display 0.96 inch I2C	1	Local display output
Breadboard	1	Circuit assembly
Jumper Wires	As required	Connections
USB Cable	1	Programming and power

Software

- Arduino IDE with the ESP32 board package
- DHT sensor library
- Adafruit SSD1306 library
- Adafruit GFX library
- ThingSpeak library and a ThingSpeak account for the cloud dashboard

Circuit Connections and Air Quality Classification

Module	Module Pin	ESP32 Pin
MQ135	AO	GPIO34
DHT22	DATA	GPIO4
OLED Display	SDA	GPIO21
OLED Display	SCL	GPIO22
All Modules	VCC	3.3V or 5V
All Modules	GND	GND

The voltage rating of each sensor module must be checked before connecting VCC, since some modules support 5V while others work best with 3.3V.

The MQ135 gives an analog value, which the sketch classifies as follows:

MQ135 Value	Status
Less than 1000	Good
1000 to 1999	Moderate
2000 and above	Poor

These thresholds are for demonstration. Accurate AQI or PPM measurement requires proper calibration of the MQ135 sensor.

Applications and Advantages

Applications

- Indoor air quality monitoring
- Classroom environment monitoring
- Laboratory monitoring
- Small workplace safety monitoring
- Home weather and air quality tracking
- IoT mini project or academic project

Advantages

- Low-cost implementation
- Easy to build and understand
- Real-time local and remote monitoring
- Uses commonly available components
- Suitable for beginners and students
- Can be expanded with more sensors

Limitations

- The MQ135 needs calibration for accurate gas concentration measurement.

- Readings may change with sensor quality and environmental conditions.
- An internet connection is required for cloud dashboard updates.
- The system is not a replacement for certified industrial air quality monitoring equipment.

Future Scope

- Add a buzzer alert for poor air quality.
- Add mobile app notifications.
- Add a PM2.5, CO2 or pressure sensor.
- Store data in a custom database.
- Add battery power support.
- Control an exhaust fan automatically based on air quality.
- Add a web dashboard hosted on a local server.

Conclusion

The ESP32-based air quality and weather monitoring system gives a simple and affordable way to track environmental conditions in real time. It shows readings locally, stores a history in the cloud, and flags unhealthy air at a glance. With sensor calibration and the planned extensions, it can grow from a learning project into a more capable monitoring tool.